A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

`prototype/ecm` · runs in `log/okuda_ECM` · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets × fields × operators × schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity	set	representation	principal operators
epithelium	`vertex`, `cell`	3D active vertex model	`shape_energy_3d`, `morphogen_growth_3d`, `d`
junction	relation on the mesh	half-edges, keyed state	`junction_myosin`, `reconnect_t1_3d`
basement membrane	`basement_membrane_particle`	MPM + explicit bonds	`..._seed/bond/remodel/bond_break`, `integ`
interstitial ECM	`mpm_particle`	MPM fibres	`ecm_seed`, `ecm_stress`, `cell_to_ecm`
(field)	`mpm_grid`	MLS-MPM background grid	`mpm_scatter/gather/grid_update`

Every particle body scatters into and gathers from the *same* grid; that sharing *is* the mechanical coupling between them, so no pairwise contact model is written. Operators carry one of seven kinds. The four dynamics kinds return a per-set delta the engine integrates once per tick; `field`, `rewire` and `structural` mutate the grid, the relation E, or the membership $|S|$ in place and *return nothing.* That last fact drives a design decision in §5.

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f's volume is the wedge from the tissue centre, $v_f = \frac{1}{3}\,\mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E \;=\; \sum_f \Big[K_A\,(A_f - A_f^0)^2 + K_P\,(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V\,(v_f - v_f^0)^2\Big] \;+\; \Lambda \sum_e m_e\,\ell_e \;+\; K_R \sum_i \big(|\mathbf{x}_i| - R_0\big)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta\mu\,\nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f,

$$s_f \leftarrow s_f\big(1 + \lambda(\rho + \mathrm{Hill}(a_f))\big), \qquad A_f^0 = A_f^{0,\mathrm{init}} s_f^2, \quad P_f^0 = P_f^{0,\mathrm{init}} s_f, \quad v_f^0 = v_f^{0,\mathrm{init}} s_f^3, \quad (2)$$

and `divide_3d` splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \to 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is material points on a background grid, fixed-corotated with $F = RS$:

$$\boldsymbol{\tau} \;=\; 2\mu\,(F - R)F^\top + \lambda J(J-1)\,I, \qquad \boldsymbol{\sigma} \;=\; \boldsymbol{\tau}/J, \qquad \sigma_{\mathrm{vM}} = \sqrt{\tfrac{3}{2}\,\mathbf{s}\!:\!\mathbf{s}}, \;\; \mathbf{s} = \boldsymbol{\sigma} - \tfrac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})I. \quad (3)$$

$\boldsymbol{\tau}$ was already computed every substep to build the affine momentum matrix and then discarded; `mpm_scatter` now optionally caches the Cauchy stress $\boldsymbol{\sigma}$ to a per-particle buffer (`store_stress`, off by default, byte-identical when off). This matters quantitatively: colouring by the volumetric measure $|J-1|$ put 0.6% of the matrix above the display floor, whereas σ_{vM} of the *same* run puts 64% there. A matrix pushed by a growing sphere *shears*; the fixed-corotated law resists volume change far more stiffly than shape change, so the volumetric measure was blind to the deformation, not the mechanics.

4. Spheroid ↔ ECM

The two solvers cannot share a world, and the reason is not a software limit. `mpm_grid` is hard-coded to $[0,1]^3$ with $dx = 1/n_{\mathrm{grid}}$, while the epithelium lives in a 50-unit box seeding cells at radius 5. Rescaling the tissue into the unit box was tried and measured, not assumed: the same run gave $200 \to 319$ cells at its own scale, $200 \to 212$ under a pure geometric rescale (it *collapses*), and $200 \to 201$ under a dimensional one—stable, but no longer growing. The cause is that the terms of Eq. (1) carry different powers of length, $K_A(\Delta A)^2 \sim s^4$, $K_P(\Delta P)^2 \sim s^2$, $K_V(\Delta V)^2 \sim s^6$, $\Lambda P \sim s$, so a 50× shrink changes *which term wins* and surface tension takes over; correcting every exponent stops the collapse and still does not restore growth. Calibrating a vertex model to a new scale is a project, and doing it badly would mean running the ECM experiment against a tissue that is no longer the reference one while the movie looked right. So the two are run as two passes joined by a single geometric scale s, applied to a *recorded* shape rather than to a simulation. The tissue surface is recorded as an angular radius map $R(\theta,\phi,t)$ over its apical vertices. Contact is a one-sided penalty plus a hard projection,

$$\mathbf{a}_i = k\,\big(R(\hat{\mathbf{u}}_i) - r_i\big)\,\hat{\mathbf{u}}_i \;\text{ for }\; r_i < R, \qquad \text{then} \qquad \mathbf{x}_i \leftarrow \mathbf{c} + \hat{\mathbf{u}}_i\,R(1+\varepsilon) \;\text{ if still inside,} \quad (4)$$

the projection (`cell_exclude_3d`) being necessary because a penalty punishes penetration rather than preventing it. The reaction is accumulated by direction into a pressure map and normalised by the solid angle each bin subtends, $P(\theta,\phi,t) = \sum_{i\in\text{bin}} k\,\text{depth}_i\,/\,(R^2\,d\Omega)$, which closes the loop back onto the cells: `ecm_growth_gate_3d` slows the cell cycle where the matrix presses,

$$g \;=\; f_0 + \frac{1-f_0}{1+(P/P_{1/2})^n}, \qquad v_f^0 \leftarrow v_f^{0,\text{prev}} + g\,\big(v_f^{0,\text{now}} - v_f^{0,\text{prev}}\big), \tag{5}$$

i.e. the growth *increment* of Eq. (2) is gated, so a loaded cell reaches its division threshold later and divides less often. This is a rate effect and compounds: it turns a 2.75× stress anisotropy into a shape. Measured, against a round control at oblateness 1.009: gating on a polar stress pattern gives 1.32–1.43, while a *flat* pattern of the same strength shrinks the tissue just as hard and leaves it round (1.015). The shape comes from the pattern, not the amount. The binding constraint proved to be *when* the pattern exists, not how strong it is: five interventions that raised the stress (stiffer matrix, denser matrix, a trapping plane, an earlier gate, plates in contact) each *lowered* the shape, and only an oblate cavity—which makes the pattern directional from first contact, 14.7 against 1.5 at frame 250—reached the ideal.

5. The two new levels

Junction myosin. Actomyosin previously existed only as the scalar Λ of Eq. (1). It is now per-junction, recruited by tension and relaxing over τ_m:

$$m_e^{\text{ss}} \;=\; \alpha\Big(1+\beta\big(\ell_e/\langle\ell\rangle - 1\big)\Big), \qquad \frac{dm_e}{dt} \;=\; \frac{m_e^{\text{ss}} - m_e}{\tau_m}, \tag{6}$$

with α the global activity—the *in silico* myosin inhibition. Because `rewire` and `structural` operators return no delta (§1), any state living on junctions is silently invalidated by every T1, division and death. So m_e is keyed by *topological identity*, the unordered vertex pair $(\min(v_i,v_j),\max(v_i,v_j))$: a surviving edge finds its own value, a new edge takes m_{new}, a deleted edge stops being asked, and `reconnect_t1_3d` and `divide_3d` need no edits. A test battery confirms $\alpha{=}1, \beta{=}0$ is bit-identical to no operator, that the radius falls monotonically over $\alpha \in [0,3]$ ($7.22 \to 6.14$), and that new junctions appear at 10.5 per division rather than at the edge count per frame—the signature that the keying holds.

Basement membrane. A monolayer of material points seeded on $R(\hat{\mathbf{u}},0)+\delta$ and joined by explicit *crosslinks*, so that the sheet can be *defective*: MPM particles are independent material points coupled only through the grid, so an MPM-only membrane has no connectivity, cannot be crosslinked and cannot fragment. A crosslink is a two-body bond (i,j) between neighbouring membrane particles, built once by a radius search on the seeded shell and carrying its own rest length ℓ_{ij}^0—it stands for the collagen IV network that carries the load (laminin and nidogen contribute little to stiffness). Adhesion is a different object entirely: a *one-body* spring between a membrane particle and a moving point on the epithelial surface. Four terms:

$$\text{crosslink:}\quad \mathbf{f}_{ij} = k_b\,(\ell_{ij}-\ell_{ij}^0)\,\hat{\mathbf{d}}_{ij}, \qquad \text{turnover:}\quad \ell_{ij}^0 \leftarrow \ell_{ij}^0 + \frac{\ell_{ij}-\ell_{ij}^0}{\tau_r}, \qquad \text{failure:}\quad \frac{\ell_{ij}}{\ell_{ij}^0} - 1 > \varepsilon_b \Rightarrow \text{bond removed}, \tag{7}$$

$$\text{integrin:}\quad \ddot{\mathbf{x}}_i = k_a\big(\mathbf{a}_i(t)-\mathbf{x}_i\big) - c_d\,\dot{\mathbf{x}}_i, \qquad \mathbf{a}_i(t) = \mathbf{c} + \hat{\mathbf{u}}_i\big(R(\hat{\mathbf{u}}_i,t)+\delta\big), \qquad c_d = 2\sqrt{k_a}\ (\zeta=1). \tag{8}$$

Do the two matrix layers interact? Only by pushing, and only as far as the grid resolves. Both particle bodies scatter into the one `mpm_grid` and gather back from it (`mpm_scatter[accumulate]`), so they exchange momentum—but no bond joins a membrane particle to a fibre, and the two layers slide past one another freely. The missing object is collagen VII: anchoring fibrils staple the basement membrane to the interstitial matrix, and their operator (`basement_to_ecm`) is designed and not written. The quantitative caveat is sharper. At $n_{\text{grid}} = 48$ the cell size is $dx = 0.021$, against a membrane thickness of 0.002 and an in-plane particle spacing of ≈ 0.005: one grid cell holds some sixteen membrane particles and is ten times thicker than the sheet. The grid does not resolve the membrane, it sees a smeared density, and any fibre within one cell of it is effectively co-moving. The strength of the membrane–matrix coupling is therefore set by dx rather than by any parameter of the model, and no membrane–matrix result should be read off this configuration without a resolution study.

Bond removal is registered `rewire`: fragmentation *is* a change of E. What is reported is not the bond count but the largest connected component—a sheet that has lost a third of its crosslinks and is still one piece is not fragmented.

The anchor $\mathbf{a}_i$ of Eq. (8) deserves stating precisely, because it is what distinguishes adhesion from contact. $\hat{\mathbf{u}}_i$ is the particle's *own* direction, frozen at seeding, so the anchor sits at a fixed *angular* position and the spring resists tangential sliding—anchoring instead to the nearest surface point would let the particle slide freely, which is contact, and the shared grid already provides that. The radial coordinate $R(\hat{\mathbf{u}}_i, t)$ follows the epithelium outward, so the bond is radially compliant and tangentially stiff, as an integrin–focal adhesion complex is. Two consequences follow. Geometrically, a particle pinned to a fixed angular position on a surface whose radius triples must accommodate area growing as R^2, so its crosslinks stretch by $\sim R$: this is the loading a basement membrane experiences under tissue growth, and without it the sheet merely slides and its bonds never strain at all (measured: mean strain 0.0000 at every bond stiffness). Mechanically, because the tissue is a replay, $\mathbf{a}_i$ is a *prescribed kinematic target* and not a mutual force pair—the epithelium feels no reaction. That is the one-way coupling of §4 again, and it bites hardest here, since integrin adhesion is precisely the route by which a stiff membrane would restrain a growing acinus.

Two calibration facts, both found by tests rather than by inspection. (i) Writing the crosslink force as $k\,\varepsilon = k(\ell - \ell^0)/\ell^0$ makes it $1/\ell^0 \approx 450\times$ stiffer than the number supplied, and the sheet tore itself apart in 40 frames; Eq. (7) is Hookean in the *extension*, with the explicit-integration bound $\Delta t_{\text{sub}} < 2/\sqrt{k_b}$. (ii) The anchor in Eq. (8) *moves*, at $v_a \approx 0.135$ box/time. An undamped spring does not track a ramp, it oscillates about it with amplitude $v_a/\omega = 9.5 \times 10^{-4}$, against a bond rest length of 2.2×10^{-3}: 43% strain oscillation on every particle, and 66 680 of 70 129 crosslinks broken. At $\zeta = 1$ the steady lag $2v_a/\omega$ is larger but *coherent* between neighbours, so it does not stretch bonds—0 broken, the sheet intact at 10% strain.

A result falls out of Eq. (7) without τ_r: sweeping ε_b over 0.05/0.20/0.60 destroyed the sheet at every value, because the surface grows 66% in linear strain by frame 150 and 260% by the end. A purely elastic basement membrane *cannot* enclose a growing epithelium; remodelling is not optional, and fragmentation becomes a race between τ_r and growth.

6. Status and the next operator

63 runs in `log/okuda_ECM`; `prototype/ecm/RUNS.md` records which are mutually comparable, since three quantities were redefined mid-campaign (notably $|J-1| \to \sigma_{\text{vM}}$, $\sim 100\times$ in "strained fraction"). The coupling of §4 is one-way; `ecm_load_3d` is the staggered two-way scheme, implemented and uncalibrated. `cell_polarity` is absent—the epithelium has no simulated basal side, its cells being wedges from the centre and the monolayer a rendering convention—as are `cell_extrude_3d` and MMP degradation.

The clearest gap is in Eq. (7). With N fixed on a shell whose radius grows 3.6×, the areal density $N/4\pi R^2$ falls 13× and every rest length must creep to 3.6× its original span: what τ_r delivers is therefore *dilution with amnesia*, not remodelling, and it is why the surviving sheet sits permanently near 10% strain. A basement membrane does not thin thirteenfold as an acinus grows—the epithelium secretes laminin and collagen IV continuously and thickness is held roughly constant. The missing operator is `basement_membrane_secrete`, kind `structural`: insert particles where areal density has fallen below target and bond them in, the sheet's analogue of `divide_3d`. Plexus supports it directly through `Level.free_slots`/`spawn` with an oversized reservoir. Only then does "the membrane survived" mean what it means biologically—that the tissue did not outgrow the membrane it was building—rather than that its crosslinks were stretched and told to forget.

1 Two things the implementation gets wrong, stated plainly

The anchor sets where the sheet sits, and only a narrow band of it works. With critical damping the sheet stopped tearing, but it then sank *through* the epithelium: the gap to the recorded surface ran $+0.0040 \to -0.0101 \to -0.0255$, every particle inside by mid-run. Tracking lag does not explain it ($2v_a/\sqrt{k} = 0.0019$, sixteen times too small). Nor—and this corrects the obvious guess—does hoop tension: quartering k_{bond} from 2×10^5 to 5×10^4 at fixed k_{adh} moves the gap by under 5% (+0.00185/+0.00183/+0.00176) and leaves the strain alone. The reason is in Eq. (7)'s companion: remodelling fixes the steady-state strain at $\dot{\epsilon}\,\tau = 0.0027 \times 60 = 0.16$ (measured 0.151–0.160 in every run, at every stiffness), so rest lengths adapt and bond force never becomes large. What sets the gap is k_{adh} against a residual inward force; the untested candidate is the interstitial matrix pressing through the shared grid, which would be the one real matrix-on-membrane effect in the model.

The sweep is narrow. At 200 frames $k_{\text{adh}} = 4 \times 10^4/6 \times 10^4/8 \times 10^4$ gives gaps $-0.0037/-0.0013/+0.0019$; 1×10^5 *reverses* to -0.069 and 2×10^5 tears 94% of the crosslinks. A stiffer spring cannot hold less well, so the reversal is a stability threshold rather than mechanics, and 8×10^4 is the usable value—at which 46% of particles are still inside. A membrane pinned at fixed angular positions on a surface that triples in radius is being asked to cover nine times the area with the same particles; the operator that would resolve this, and which the model does not have, is `basement_membrane_secrete`.

`activity` is degenerate with Λ. Actomyosin entered Eq. (1) as the single scalar Λ and now also as a per-junction variable, and the two do not double-count: myosin *multiplies* the line term, $\Lambda \sum_e m_e \ell_e$, and the recruitment setpoint of Eq. (6) is written against the *running mean* edge length $\langle \ell \rangle$, so

$$m_e^{ss} = a\Big(1 + \beta\big(\ell_e/\langle \ell \rangle - 1\big)\Big) \quad \Longrightarrow \quad \langle m^{ss} \rangle = a.$$

At $a = 1$ the mean is pinned to one, Λ keeps the meaning it was calibrated with, and myosin is a pure zero-mean redistribution—which is why the ablation is bit-identical rather than merely close. But it follows that a, which scales every junction uniformly, is *not identifiable against* Λ: the monotone radius response over $a \in [0, 3]$ ($7.216 \to 6.138$) is reproduced by sweeping Λ and is a consistency check, not a result. The new parameters are β, τ and m_{new}.

The epithelial surface is not an entity. It is $R(\hat{\mathbf{u}}, t)$, a 32×64 angular table, read by four operators (`cell_to_ecm`, `cell_exclude_3d`, `integrin_adhesion`, `basement_membrane_seed`). Being a table and not a Level it carries no state and can receive no delta, which is the structural reason the adhesion of Eq. (8) is one-way. The Plexus2 form is a Level `surface`, one member per outer face with $(\mathbf{x}, \mathbf{v}, \hat{\mathbf{n}}, A)$, written each frame by a `broadcast` from the replayed mesh; the four operators then become ordinary `lateral` ones between two Levels, the angular lookup disappears, and—because a Level can be integrated—the integrin spring becomes mutual. A stronger version is available once the sheet is anchored, since the membrane particles are themselves a discretisation of that surface and could carry the contact directly; but runs without a membrane still need a contact path, so `surface` is the general answer and membrane-as-surface the specialisation.